EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 5 – Lecture 1

Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **AB-227**

Number Systems

Decimal number system: It is a base-10 counting because the total number of symbols used is ten. When we represent a number, say 752 and 907, the positional values of the numbers and their decimal equivalent would be. A decimal number of, say, 752 is represented as

Decimal $752 = 7 \times 10^2 + 5 \times 10^1 + 2 \times 10^0$

A decimal number of say, 824.48 is represented as

Decimal $824.48 = 8 \times 10^2 + 2 \times 10^1 + 4 \times 10^0 . 4 \times 10^{-1} + 8 \times 10^{-2}$

Number	Value	Position	Positional value	Decimal equivalent
7 5 2				
	2	0	10^0	$2 \times 10^0 = 2$
	5	1	10^1	$5 \times 10^1 = 50$
	7	2	10^2	$7 \times 10^2 = 700$
				752

The multiples of 10 used are called weighted values. The decimal number 824.48 is represented as $(824.24)_{10}$ as 10 is the base.

Binary system: Here, a number is expressed by two symbols or digits, 0 and 1. Since the total number of symbols or digits used is two, it has a base 2. The binary symbols 0 and 1 are called binary digits or bits. A group of eight bits is called a byte. In this system the positional value is used in the same way as the decimal system. For example, two binary numbers, 11010 and 110011 are represented as their decimal equivalent as

Number	Value	Position	Positional value	Decimal equivalent
1 1 0 1 0				
	0	0	2^0	$0 \times 2^0 = 0$
	1	1	2^1	$1 \times 2^1 = 2$
	0	2	2^2	$0 \times 2^2 = 0$
	1	3	2^3	$1 \times 2^3 = 8$
	1	4	2^4	$1 \times 2^4 = 16$
				26

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Number Systems

Conversion of Binary to Decimal: A binary number can be converted into its decimal equivalent number by noting that successive digits from the extreme right of a binary number are the coefficients of ascending power of 2 beginning with zeroth power of 2.

To convert the binary number 1100101 into its decimal equivalent:

1	1	0	0	1	0	1
1×2^6	1×2^5	0×2^4	0×2^3	1×2^2	0×2^1	1×2^0

$64 + 32 + 0 + 0 + 4 + 0 + 1 = 101$
Decimal number is 101. Therefore, $(1100101)_2 = (101)_{10}$

To convert fractional binary number to fractional decimal number, the decimal equivalent of the binary fraction is determined by multiplying each digit in the fraction by 2^{-1}, 2^{-2}, 2^{-3}, 2^{-4}, . . . etc., beginning with the first digit after the binary point. The products are then added to get the decimal fraction.

To convert the fractional binary number 0.1011 into decimal fraction:

$1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} + 1 \times 2^{-4}$
$= 0.5 + 0 + 0.125 + 0.0625$
$= 0.6875$

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Number Systems

Conversion of Decimal to Binary: To convert a decimal number into a binary equivalent number, the decimal number is expressed as a sum of ascending power of 2. Suppose we are to find the binary equivalent of the decimal number 45. The number 45 is arrived at by adding 32 + 8 + 4 + 1. The 1s and 0s are placed as shown

2^5		2^4		2^3		2^2		2^1		2^0	
32	+	0	+	8	+	4	+	0	+	1	= 45

The binary number is 101101

Alternatively, we can divide the decimal number progressively by 2 till the quotient is zero. The remainders of the successive divisions, expressed in the reverse order, gives the binary equivalent number. To convert decimal 23 into its binary equivalent number:

			Remainder
Divide	23 by 2	$2\overline{)\begin{matrix}23\\11\end{matrix}}$	1
Divide	11 by 2	$2\overline{)\begin{matrix}11\\5\end{matrix}}$	1
Divide	5 by 2	$2\overline{)\begin{matrix}5\\2\end{matrix}}$	1
Divide	2 by 2	$2\overline{)\begin{matrix}2\\1\end{matrix}}$	0
Divide	1 by 2	$2\overline{)\begin{matrix}1\\0\end{matrix}}$	1

Arranging the remainders in the reverse order, i.e., from bottom upwards, the binary equivalent number is 10111

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To convert fractional decimal number to fractional binary number, first, multiply the decimal fraction by 2. The result contains an integer part and a fractional part. Then, separate the integer part and write it in a column. The fractional part becomes the new fraction. Repeat these steps till the fractional part becomes zero or the desired number of binary places are obtained. The 1s and 0s separated placed together gives the required fractional binary number.

To convert 0.8125 into its binary fraction:

		Fractional part	Integer part
0.8125×2	$= 1.6250$	0.6250	1
0.6250×2	$= 1.2500$	0.2500	1
0.2500×2	$= 0.5000$	0.5000	0
0.5000×2	$= 1.0000$	0	1

The binary decimal number is 0.1101

To convert 0.65 into its binary fraction:

		Fractional pat	Integer part
0.65×2	$= 1.3$	0.3	1
0.3×2	$= 0.6$	0.6	0
0.6×2	$= 1.2$	0.2	1
0.2×2	$= 0.4$	0.4	0
0.4×2	$= 0.8$	0.8	0
0.8×2	$= 1.6$	0.6	1
0.6×2	$= 1.2$	0.2	1

In the last step we get back 0.2 as the fractional part. To get the approximate result we can terminate the process at this stage. Thus, the approximate binary fraction is 0.1010011.

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Binary addition: The binary addition is performed by using the following rules:

(i) 0 + 0 = 0
(ii) 1 + 0 = 1
(iii) 0 + 1 = 1
(iv) 1 + 1 = 10 (since, decimal equivalent of 10 is 2)

Example:

$$\begin{array}{r} 10001 (= 17) \\ +111 (= 7) \\ \hline 11000 (= 24) \end{array}$$

First column 1 + 1 = 10, place 0 and carry 1
Second column, 1 + 0 = 1, 1 + 1 = 10, place 0 and carry 1
Third column, 1 + 0 = 1, 1 + 1 = 10, place 0 and carry 1
Fourth column, 1 + 0 = 1, place 1 and carry 0
Fifth column, 1 + 0 = 1, place 1

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Binary subtraction: The binary subtraction is performed by using the following rules:

(i) 0 – 0 = 0
(ii) 1 – 0 = 1
(iii) 1 – 1 = 0
(iv) 10 – 1 = 1 (decimal equivalent of 10 is binary 2)

Example:

$$\begin{array}{r} 110\,(=6) \\ -11\,(=3) \\ \hline 011\,(=3) \end{array}$$

The first column is 10 – 1 = 1 (after borrowing 1 from the second column). The borrow is carried and added to the lower number of second column to get 1 + 1 = 10. So, the second column gives 1 – 0 = 1 and the third column gives 1 – 1 = 0.

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Binary multiplication: Binary multiplication is carried out using the following rules:

(i) $0 \times 0 = 0$
(ii) $1 \times 0 = 0$
(iii) $0 \times 1 = 0$
(iv) $1 \times 1 = 1$

Example:

			1	1	1	(= 7)
			1	0	1	(= 5)
			1	1	1	
		0	0	0	×	
	1	1	1	×	×	
1	0	0	0	1	1	(= 35)
	1 + 1	1 + 1	1 + 1	1 + 0	1 + 0	
	= 10	= 10	= 10	= 1	= 1	
		carry 1	carry 1			

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Octal number system: Octal system has a base of 8 counting from 0 to 7. To convert a **decimal number into octal equivalent** of decimal number we divide the integer repeatedly by 8 till the quotient is zero. The remainders written in reverse order gives the octal equivalent number. For example, let us find the octal equivalent of decimal number 756.

$756 \div 8 = 94$	Remainder = 4
$94 \div 8 = 11$	Remainder = 6
$11 \div 8 = 1$	Remainder = 3
$1 \div 8 = 0$	Remainder = 1

The octal equivalent of 756 is 1364.

To convert an **octal to binary** each octal symbol is replaced by a 3-bit binary equivalent

Octal	Binary
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

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Conversion of **binary number into an equivalent octal number** can be done in the following way: Split the binary digits into groups of three digits, each beginning from the right end for integers and from the left end for fractions. Replace each group by its equivalent octal number.

For example, to convert binary 11001 into octal equivalent, the binary number is grouped as 011 001. Replacing each group by its octal equivalent we get the number 31.

To convert binary 0.111 001110 into its equivalent octal, the binary number is grouped into 3-bit form as 0.111 001 110. The equivalent is 0.716.

To convert an **octal number into its decimal equivalent** the following steps be followed: Consider the octal number from the extreme right as the coefficient of the ascending power of 8, the starting power of 8 at the extreme right has to be taken as zero. If the octal number has a fraction, the fractional numbers have to be considered as the coefficient of ascending power of 8^{-1}.

Example: 732.14 $= 7 \times 8^2 + 3 \times 8^1 + 2 \times 8^0 + 1 \times 8^{-1} + 4 \times 8^{-2}$

$= 448 + 24 + 2 + 0.125 + 0.0625$

$= 474.1875$

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Number Systems

Hexadecimal number system: In hexadecimal system we use 16 symbols, viz. 0 to 9 and the capital letters A B C D E F for 10 to 15. Hence, the hexadecimal number system uses a base of 16. To convert a **binary number into a hexadecimal number** (Hex) the binary digits are grouped into 4 binary bits, counting from the right end for integral numbers and from the left end for fractions. Each group is replaced by its equivalent.

Binary	Hex	Binary	Hex	Binary	Hex	Binary	Hex
0000	0	0100	4	1000	8	1100	C
0001	1	0101	5	1001	9	1101	D
0010	2	0110	6	1010	A	1110	E
0011	3	0111	7	1011	B	1111	F

To convert 1111 1110 0011 into hexadecimal equivalent, group the binary as (from right to left) 1111 1110 0011, and replace the group by their equivalent as FE3.

To convert 1111 10001.100 1100 1101 into hexadecimal equivalent, group the binary as 0001 1111 0001 . 1001 1001 1010, and replace the groups by their equivalent to get 1F1.99A

To **convert Hex number 6D5 into its binary equivalent**, starting from right we replace binary equivalent of 5, D, and 6 as

(6)	(D = 13)	(5)
0110	1101	0101

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Thus, conversion between Hex and binary is very quick. For conversion of decimal number into binary equivalent, or useful system, a binary coded decimal system, or **BCD system** is often used. It is a code in which individual decimal digits are replaced by a group of 4 binary bits in a row. The weights of the 4 bits from left to right of the binary string are 8, 4, 2, and 1, respectively. The BCD code is also called eight four two one code.

Example: To express **2534 in BCD form**.

Decimal form	2	5	3	4
BCD form	0010	0101	0011	0100

Thus, $(2534)_{10} = (0010010100110100)_{BCD}$.

It is to be noted that this system is totally different from the method of conversion of decimal numbers into binary numbers.

The binary equivalent of 2534 is calculated as

2^{11}	2^{10}	2^9	2^8	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
2048	1024	512	256	128	64	32	16	8	4	2	1
1	0	0	1	1	1	1	0	0	1	1	0

$2048 + 0 + 0 + 256 + 128 + 64 + 32 + 0 + 0 + 4 + 2 + 0 = 2534$

Therefore, $(2534)_{10} = (100111100110)_2$

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Module 5 – Lecture 2

Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **C-509**

Logic gates and their truth tables

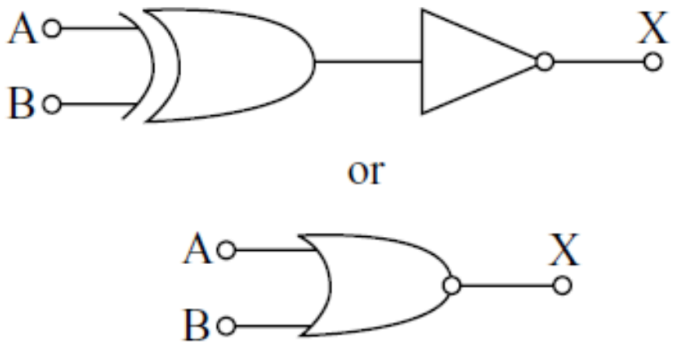
A logic gate is a simple device used to make digital integrated circuits. Diodes and transistors are used to perform switching functions in logic gates. Logic gates basically have one or more inputs and only one output. Logic gates may be classified into two categories; combinational type (with no memory) and sequential type (with memory). In digital circuits or systems there are two possible states or conditions. A switch may be closed or open, i.e., on or off. These two conditions may be represented by 1 and 0, respectively. The two logic levels, i.e., 1 and 0 represent, respectively, voltage high or voltage low.

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Gate name	Gate symbol and Boolean equation	Truth table
NOT	input x, output y; Boolean equation: $y = \overline{x}$	input x | output y 0 | 1 1 | 0
OR	inputs x, y; output z; Boolean equation: $z = x + y$	input x y | output z 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1
AND	inputs x, y; output z; Boolean equation: $z = x.y$	input x y | output z 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1
NAND	inputs x, y; output z; Boolean equation: $z = \overline{x.y}$	input x y | output z 0 0 | 1 0 1 | 1 1 0 | 1 1 1 | 0
NOR	inputs x, y; output z; Boolean equation: $z = \overline{x + y}$	input x y | output z 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 0
EXOR	inputs x, y; output z; Boolean equation: $z = x \oplus y = \overline{x}y + x\overline{y}$	input x y | output z 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 0

Logic gates and their truth tables

EX-NOR gate:



Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

$X = AB + A'B' = (A \oplus B)'$

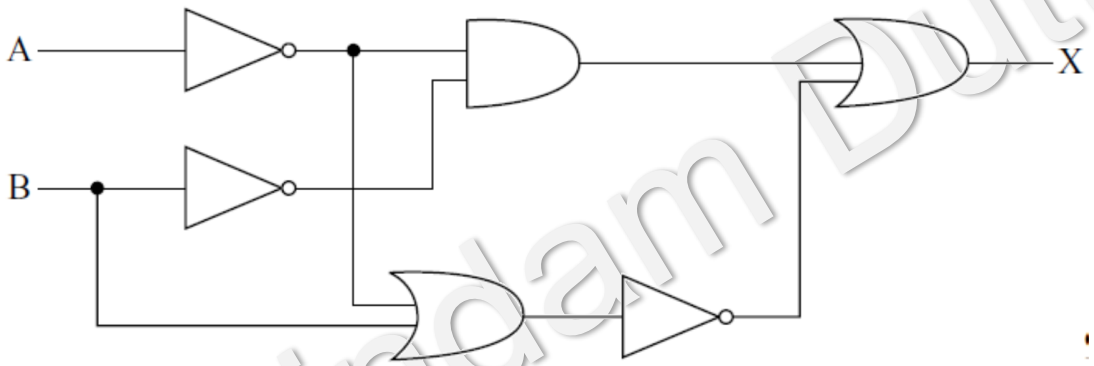
- **Three basic gates are NOT, OR, and AND gates** and the combinational gates are NAND, NOR, EXOR and EX-NOR gates.
- **NAND and NOR gates are called the universal gates** as all other gates can be created from a suitable network of just NAND or just NOR gates.
- A logic gate performs a logical operation on one or more logic inputs and produces a single logic output.
- Logic gates are primarily implemented using resistors, diodes, and transistors.

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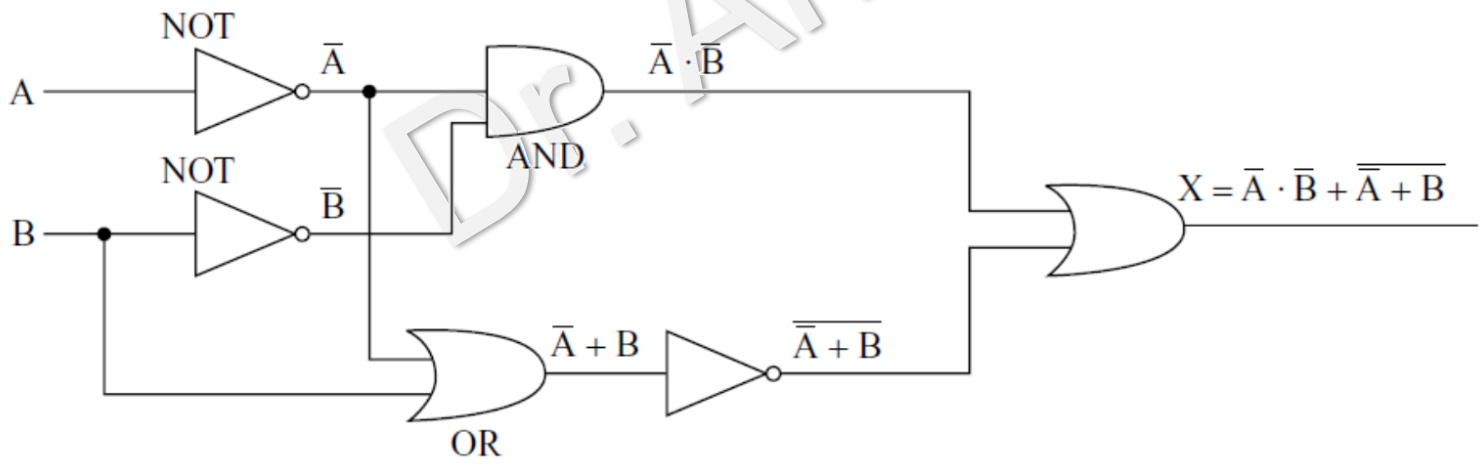
Combinational Circuits

A **combinational circuit**, also called a combinational logic circuit, is a digital electronic circuit formed using a combination of logic gates and whose output is determined by present inputs only.

Q: Write the Boolean expression and truth table for the logic gate circuit.



A: Boolean expression:



$$X = \overline{A}.\overline{B} + \overline{\overline{A} + B}$$

$$= Y + Z$$

$$Y = \overline{A}.\overline{B}$$

$$Z = \overline{\overline{A} + B}$$

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Combinational Circuits

Now let us write the truth table. We first write the various combinations of 0 and 1 for the two inputs, A and B. There are $2^N = 2^2 = 4$ combinations. For each row of inputs, we will calculate Y and Z, respectively. Then we would add Y and Z, to get X.

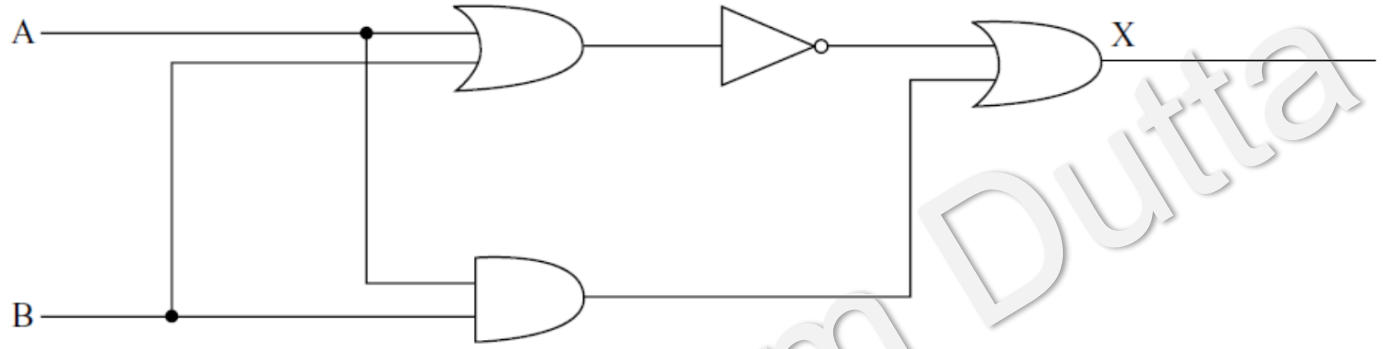
Truth Table

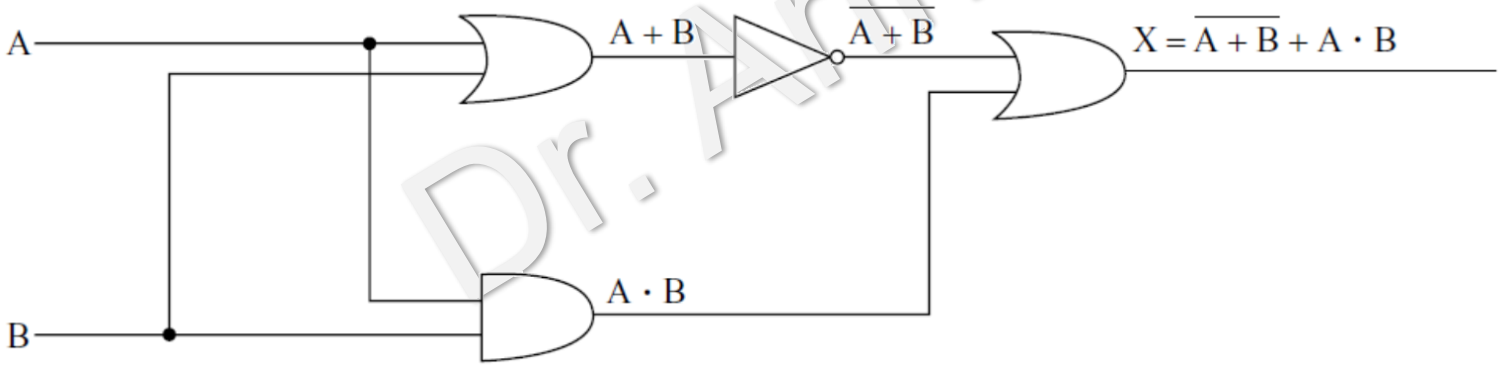
A	B	Y	Z	X
0	0	1	0	1
0	1	0	0	0
1	0	0	1	1
1	1	0	0	0

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Combinational Circuits

Q: Write the Boolean expression and construct the truth table.





$$X = \overline{A + B} + A.B$$
$$= \overline{Y + Z}$$
where $Y = A + B$ and $Z = A.B$

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Combinational Circuits

For constructing the truth table, we will first write the various combinations of the inputs at A and B in the form of binary numbers (the total number of possible combinations being 2^N where N is the number of inputs). We then calculate Y and Z, and then X is calculated.

Truth Table

A	B	Y	Z	X = Y + Z
0	0	1	0	1
0	1	0	0	0
1	0	0	0	0
1	1	0	1	1

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Combinational Circuits

Boolean expression for any logic function can be obtained from the truth table. This is done by taking the sum of all terms which correspond to all those combinations of inputs for which the output attains a high value.

Here, The first row and fourth row have to be considered where output is high, i.e., 1. Inputs A and B are to be considered 1. The complement A' and B' therefore 0.

For row 1 $Y = \overline{A + B}$

For row 4 $Z = A.B$

The Boolean expression for the output is $X = Y + Z = \overline{A + B} + A.B$

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

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Combinational Circuits

Q: The truth table for three variable inputs and single output logic circuit is shown below. Note that for three inputs the combinations are $2^3 = 8$. Construct the logic circuit using AND and OR gates.

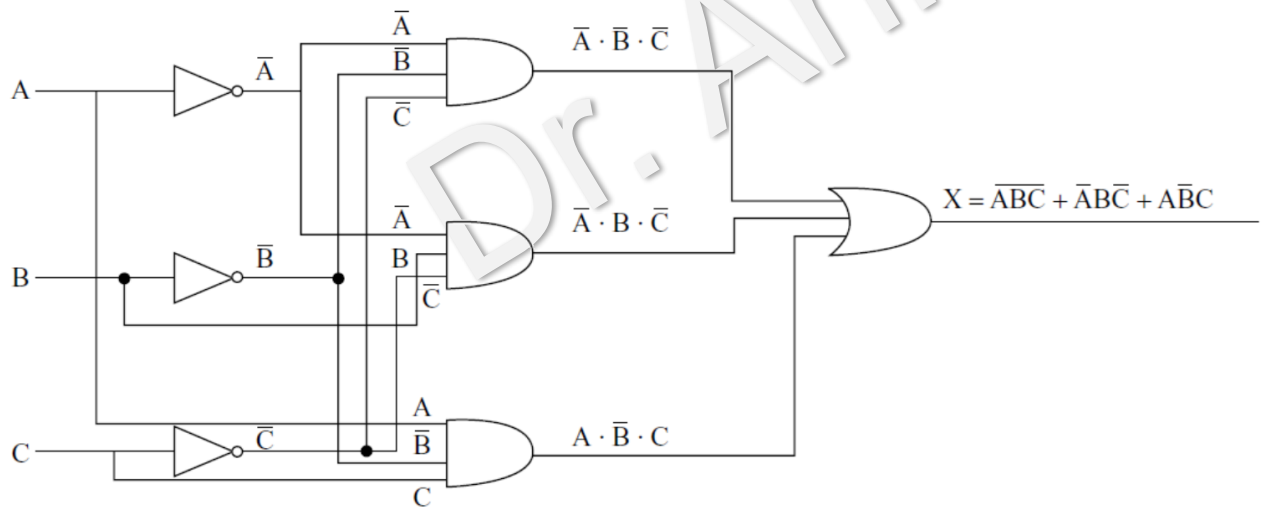
From observing the truth table we find that output is high, i.e., 1 for the inputs at rows 1, 3, and 6. Inputs at A, B, and C is 1 and complement A', B', C' is 0.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

By writing the sum of all terms for which the output attains a high or 1

$$X = \overline{ABC} + \overline{A}B\overline{C} + A\overline{B}C$$

Using the above Boolean expression, the logic circuit is drawn



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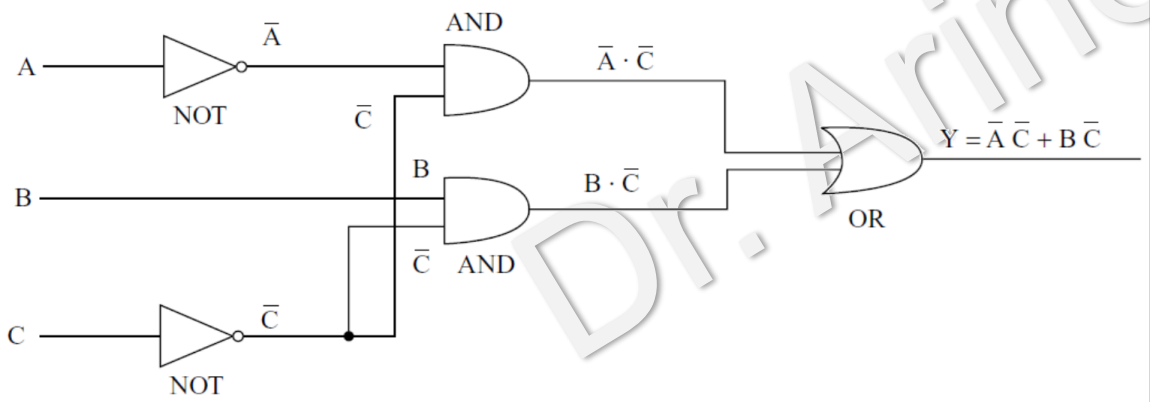
Combinational Circuits

Q: Draw the logic circuit for the Boolean expression expressed in two forms. Also write the truth table

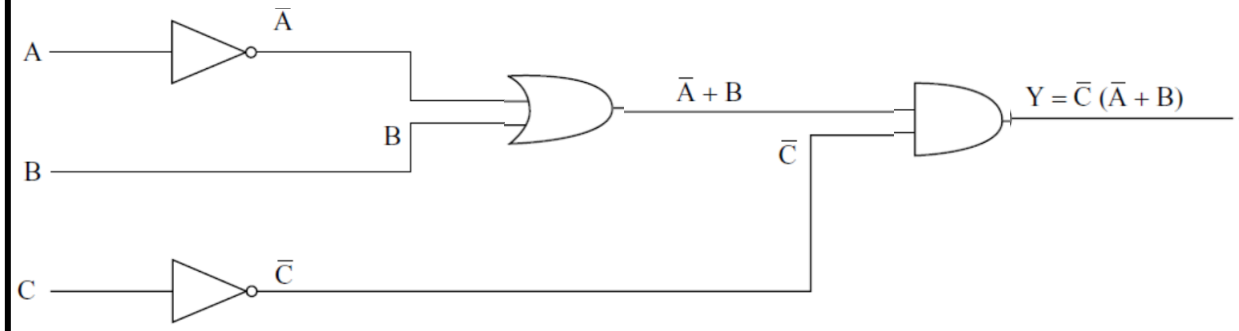
$$Y = \overline{A}\,\overline{C} + B\overline{C} \qquad \text{(i)}$$

$$Y = \overline{C}\,(\overline{A} + B) \qquad \text{(ii)}$$

For (i)



For (ii)



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Combinational Circuits

...contd.

$$Y = \overline{A}\,\overline{C} + B\overline{C}$$
$$= P + Q$$

Truth table

A	B	C	P	Q	Y
0	0	0	1	0	1
0	0	1	0	0	0
0	1	0	1	1	1
1	0	0	0	0	0
1	0	1	0	0	0
1	1	0	0	1	1
0	1	1	0	0	0
1	1	1	0	0	0

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Arithmetic Circuits

Combinational logic circuits are used to make adders and subtractors. In the central processing unit (CPU) of a computer, arithmetic functions are performed in a section called arithmetic logic unit (ALU). This section can perform functions like add, subtract, multiply, divide, compare, complement, shift, and perform logic functions like AND, OR, and XOR.

An **adder circuit** is one of the important digital circuits used in computers, calculators, digital processing units, etc. There are two types of adder circuits named **half-adder** and **full-adder**.

Half-adder: It is designed to add two binary digits. It provides the output along with a carry value (if any). The circuit is designed by connecting an EX-OR gate and one AND gate. It has two input terminals and two output terminals for sum and carry. The output of the EX-OR gate is the sum of two bits while the output of the AND gate is the carry. However, the carry obtained in one addition will not be forwarded in the next addition, so it is called half adder.

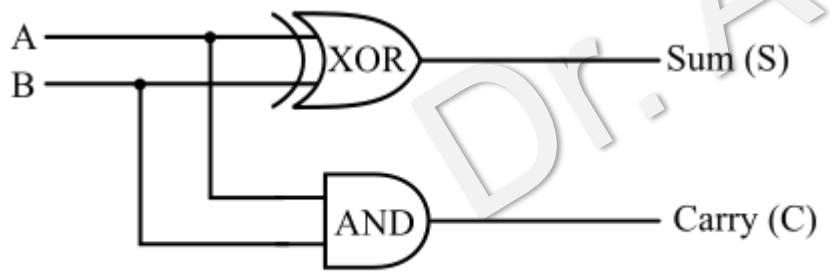


Figure - Half Adder Circuit

Input		Output	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

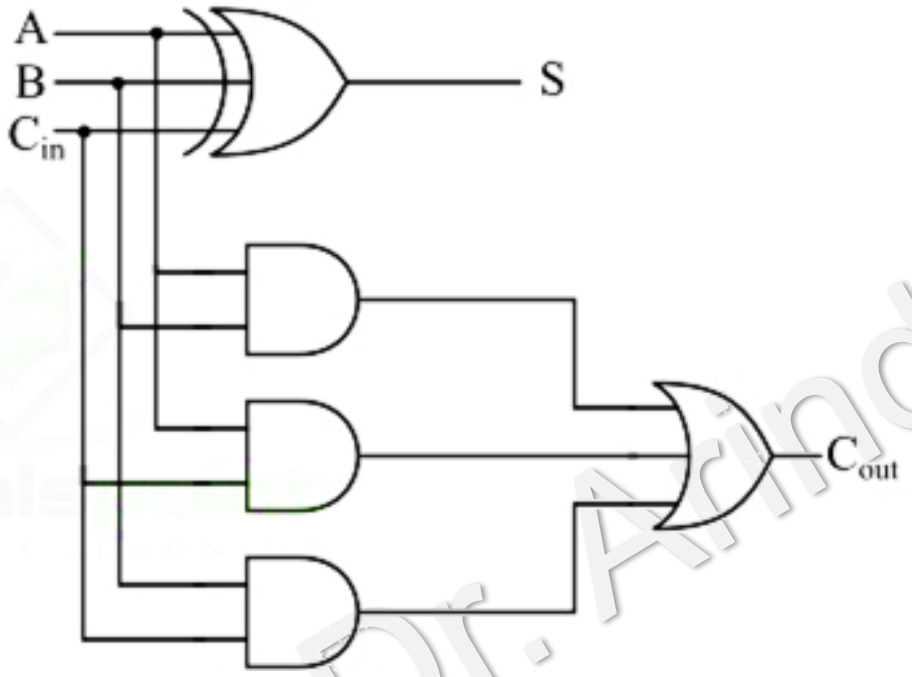
$S = A'B + AB' = A \oplus B$

$C = A \cdot B$

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Arithmetic Circuits

Full-adder: It is designed to add three binary digits and produce two outputs. Of the three binary digits, two are the inputs and one is the carry forwarded from the previous addition. The circuit of the full adder consists of one EX-OR gate, three AND gates and one OR gate.



Input			Output	
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$S = A'B'C_{in} + A'BC_{in}' + AB'C_{in}' + ABC_{in}$

$= C_{in} (A'B' + AB) + C_{in}' (A'B + AB')$

$= C_{in} (A \oplus B)' + C_{in}' (A \oplus B)$

$= A \oplus B \oplus C_{in}$

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Arithmetic Circuits

$C_{out} = A'BC_{in} + AB'C_{in} + ABC_{in}' + ABC_{in}$

$= A'BC_{in} + AB'C_{in} + AB\,(C_{in} + C_{in}')$

$= A'BC_{in} + AB'C_{in} + AB$ [Inverse Law]

$= A'BC_{in} + AB'C_{in} + AB\,(1 + C_{in})$ [Null Law]

$= A'BC_{in} + AB'C_{in} + ABC_{in} + AB$

$= BC_{in}\,(A'+A) + AB'C_{in} + AB$

$= BC_{in} + AB'C_{in} + AB$ [Inverse Law]

$= BC_{in} + AB'C_{in} + AB\,(1 + C_{in})$ [Null Law]

$= BC_{in} + AB'C_{in} + AB + ABC_{in}$

$= BC_{in} + AC_{in}\,(B'+B) + AB$

$= BC_{in} + AC_{in} + AB$ [Inverse Law]

Also, $C_{out} = A'BC_{in} + AB'C_{in} + ABC_{in}' + ABC_{in}$

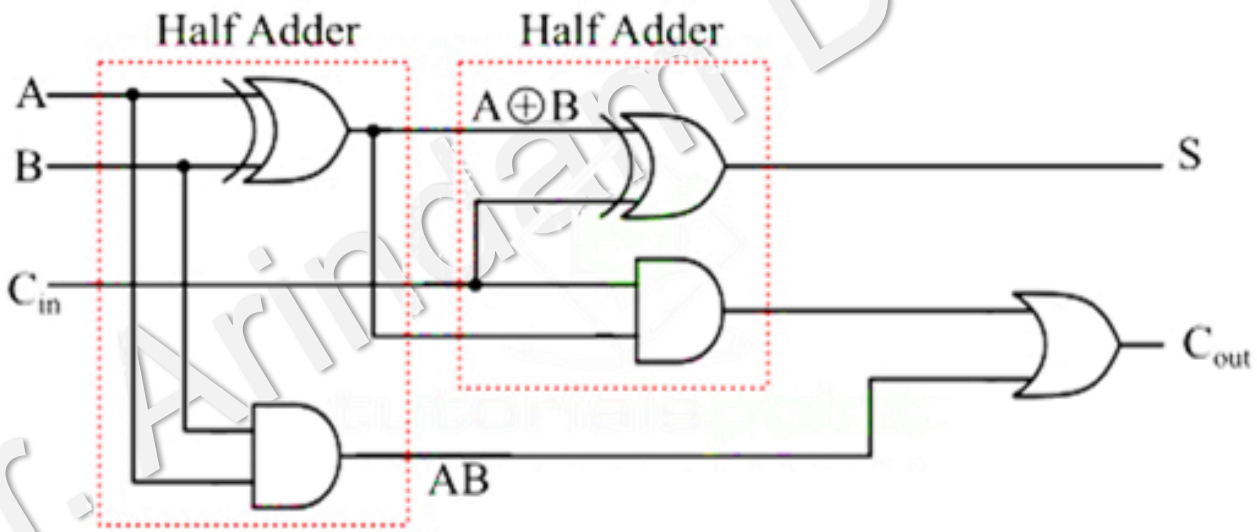
$= (A'B + AB')\,C_{in} + AB\,(C_{in} + C_{in}')$

$= (A \oplus B)\,C_{in} + AB$ [Inverse Law]

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Arithmetic Circuits

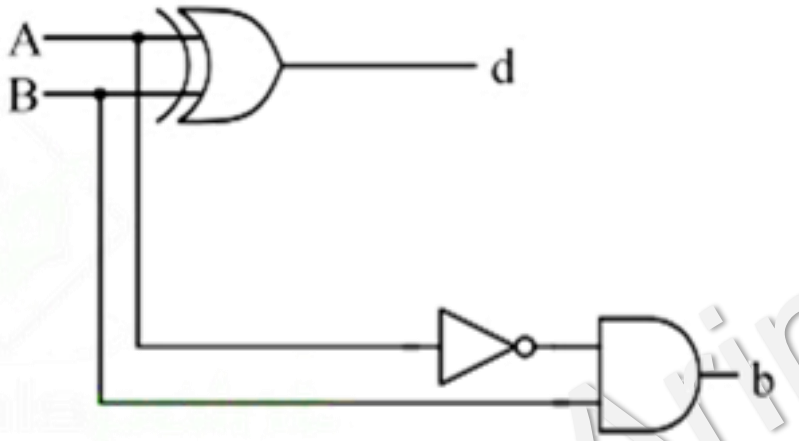
Implementation of Full Adder using Half Adder and Alternative circuit for Full Adder: Two half adder circuits can be combined to form a full-adder circuit. In this case, the circuit of the full adder consists of two EX-OR gates, two AND gates and one OR gate. The equation obtained by the EX-OR gate is the sum of the binary digits. While the output obtained by AND gate is the carry obtained by addition.



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Arithmetic Circuits

A **half-subtractor** is a combinational logic circuit that have two inputs and two outputs (i.e. difference and borrow). The half subtractor produces the difference between the two binary bits at the input and also produces a borrow output (if any). In the subtraction (A-B), A is called as Minuend bit and B is called as Subtrahend bit.



Inputs		Outputs	
A	**B**	**d (Difference)**	**b (Borrow)**
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

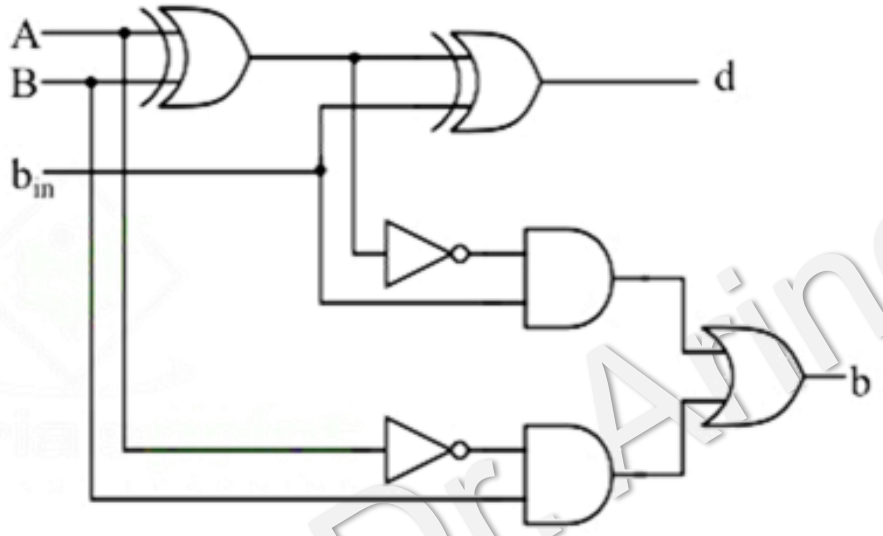
$$d = A'B + AB' = A \oplus B$$

$$b = A'B$$

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Arithmetic Circuits

A **full-subtractor** has three inputs A, B, b_{in} and two outputs d and b. Where, A is the minuend, B is subtrahend, b_{in} is borrow produced by the previous stage, d is the difference output and b is the borrow output. The half subtractor can only be used to find the difference of LSBs (Least Significant Bits) of two binary numbers. Thus, if there is any borrow during the subtraction of the LSBs, it will affect the subtraction of the next bits of numbers. To overcome this problem of the half subtractor, a full subtractor is realized.



Inputs			Outputs	
A	**B**	**b_{in}**	**d (Difference)**	**b (Borrow)**
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

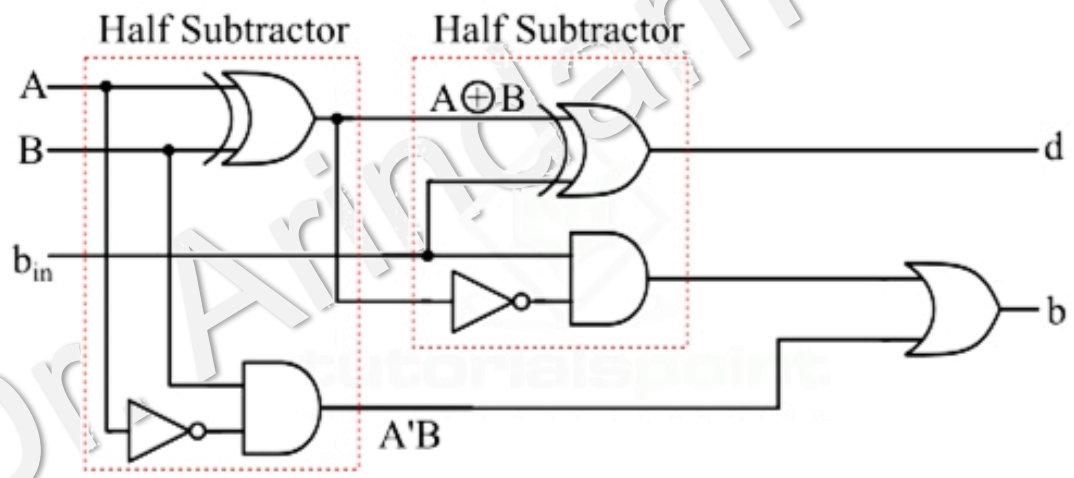
$$d = A'B'b_{in} + AB'b_{in}' + A'Bb_{in}' + ABb_{in} = (A \oplus B) \oplus b_{in}$$

$$\begin{aligned} b &= A'B'b_{in} + A'Bb_{in}' + A'Bb_{in} + ABb_{in} \\ &= A'B\,(b_{in} + b_{in}') + (AB + A'B')\,b_{in} \\ &= A'B + (A \oplus B)'\,b_{in} \end{aligned}$$

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Arithmetic Circuits

Implementation of a Full Subtractor using two Half Subtractors: A full subtractor can be realized by cascading two half subtractors. It will take two half-subtractors and one OR gate. The first half subtractor performs XOR operation on input bits A and B, and AND operation on A' and B to produce an intermediate borrow bit. The second half subtractor performs the XOR operation on the output of first XOR gate and the input borrow bit (b_{in}), and the AND gate of the second half circuit gives an output equal to (A'B + AB')'.b_{in}. The output of the second XOR gate is the output different bit (d), and the output borrow bit (b) is obtained by ORing the outputs of two AND gates.



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EEE1001
ELECTRIC CIRCUITS AND SYSTEMS

Module 5 – Lecture 3

Dr. Arindam Dutta

Assistant Professor
School of Electrical and Electronics Engineering
VIT Bhopal

Email: **arindamdutta@vitbhopal.ac.in**
Cabin: **AB-227**

Multiplexer (MUX):

- Combinational circuit that allows one of many input signals to be selected and transmitted to a single output line
- Think of a MUX like a railway junction where multiple tracks converge. The signal controls which track the train (data) will follow

Demultiplexer (DEMUX):

- Opposite of a multiplexer. It takes a single input signal and routes it to one of many output lines based on a control signal
- A DEMUX is like a dispatcher that sends a single train (data) to one of several possible tracks, depending on the signal

Applications: communication systems, data acquisition systems, computer networks, etc.

Advantages:

- Efficient data handling and processing with minimum components
- Minimized hardware reduces overall cost

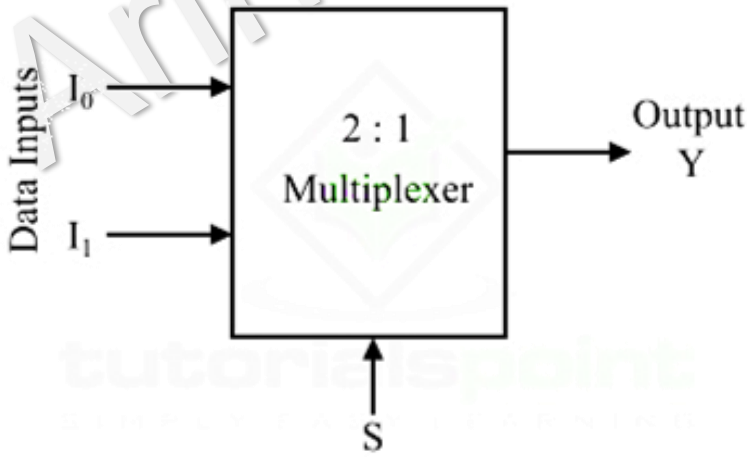
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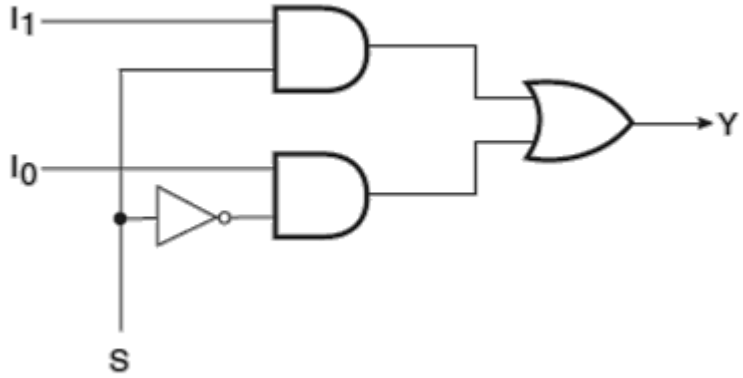
Multiplexer (MUX):

- The selection of input is determined by a set of control signals or select lines
- The number of select lines (n) determines how many inputs (2^n) can be selected
- **Control Signals:** binary signals that control which input is selected. For e.g., for 1 control lines, we can select from 2 different inputs ($2^1 = 2$)
- For a 2:1 multiplexer, if the inputs are I_0 and I_1, and the control signal is S, the output (Y) will be one of these inputs depending on S.

Truth table

S	Y
0	I_0
1	I_1





$$Y = \bar{S}I_0 + SI_1$$

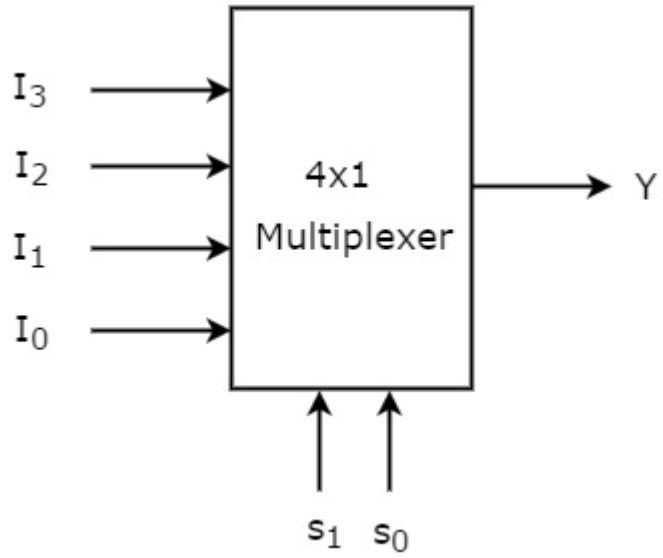
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Multiplexer (MUX):

- For 2 control lines, we can select from 4 different inputs ($2^2 = 4$)
- For a 4:1 multiplexer, if the inputs are I_0, I_1, I_2, and I_3, and the control signals are S_1 and S_0, the output (Y) will be one of these inputs depending on the combination of S_1 and S_0

Truth table

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

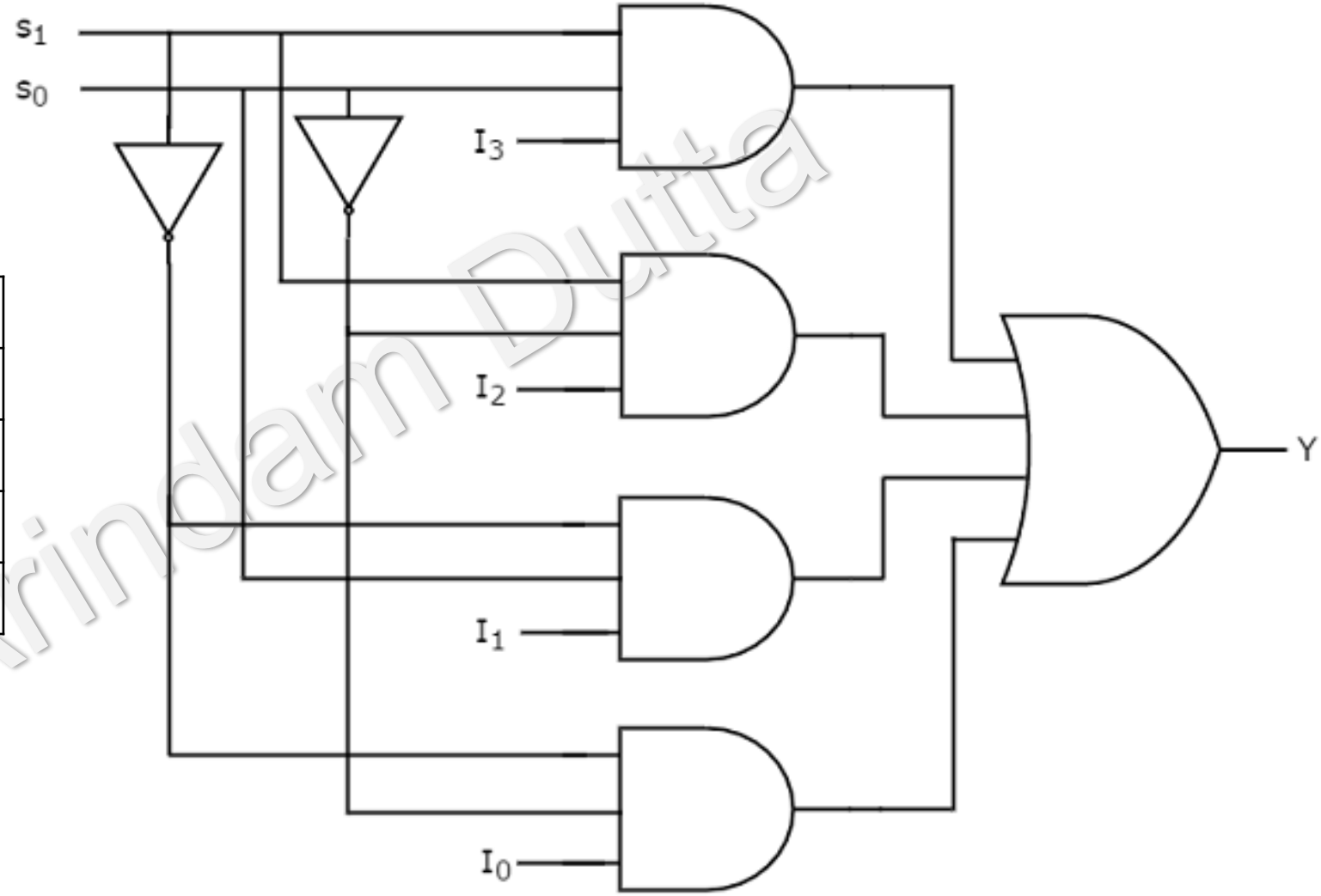


$$\mathbf{Y = \overline{S_1}\,\overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_1 \overline{S_0} I_2 + S_1 S_0 I_3}$$

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Internal logic circuit of 4:1 MUX

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



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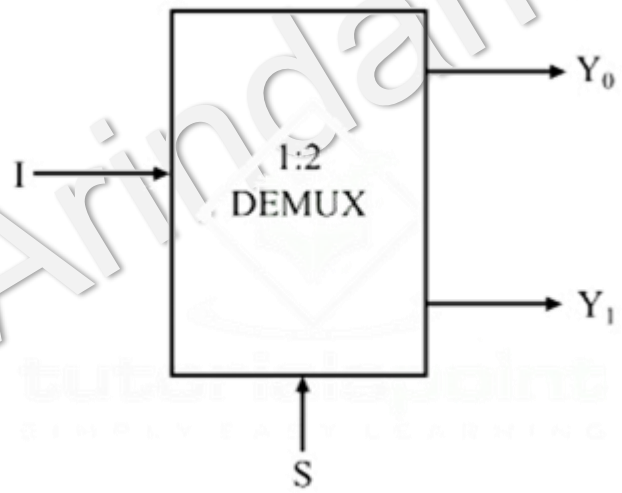
Demultiplexer (DEMUX):

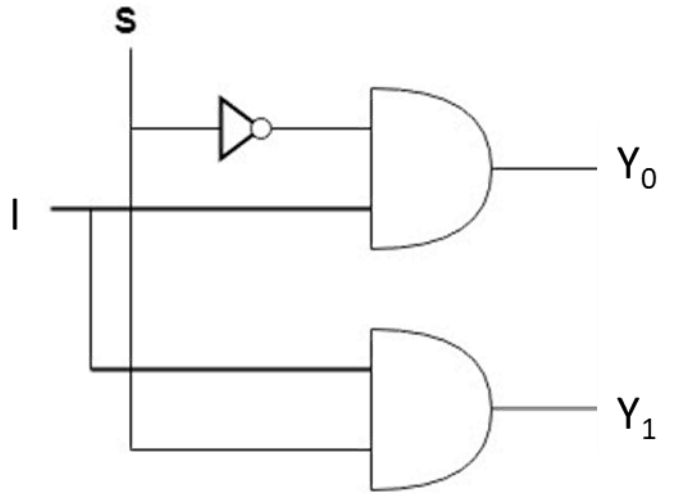
- The output line that receives the input signal is determined by the control signals or select lines
- The number of select lines (n) determines the number of outputs (2^n)
- For a 1: 2 DEMUX, for one input (I) and two outputs Y_0 and Y_1, and the control signal S, the input will be directed to one of the outputs depending on S

Truth table

S	Y_1	Y_0
0	0	I
1	I	0

$$Y_0 = \bar{S}\,I \qquad Y_1 = S\,I$$





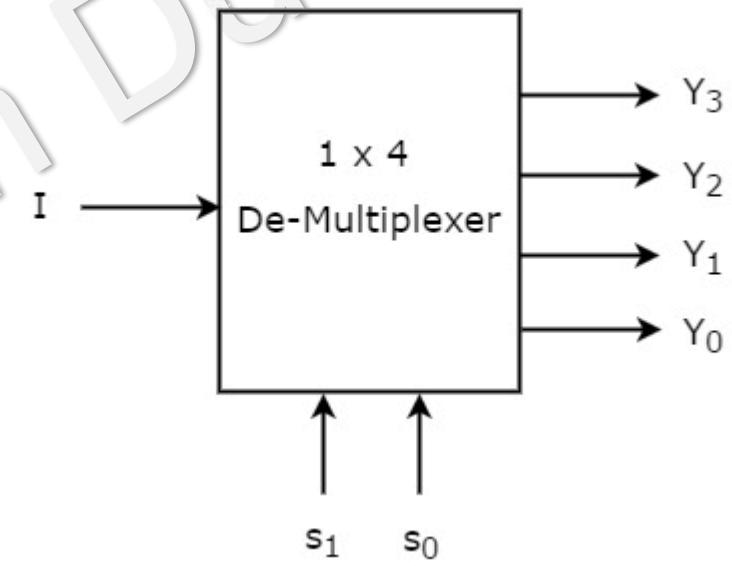
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Demultiplexer (DEMUX):

- For a 1: 4 DEMUX, for one input (I) and four outputs Y_0, Y_1, Y_2, and Y_3, and the control signals S_1 and S_0, the input will be directed to one of the outputs depending on the combination of S_1 and S_0

Truth table

S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0



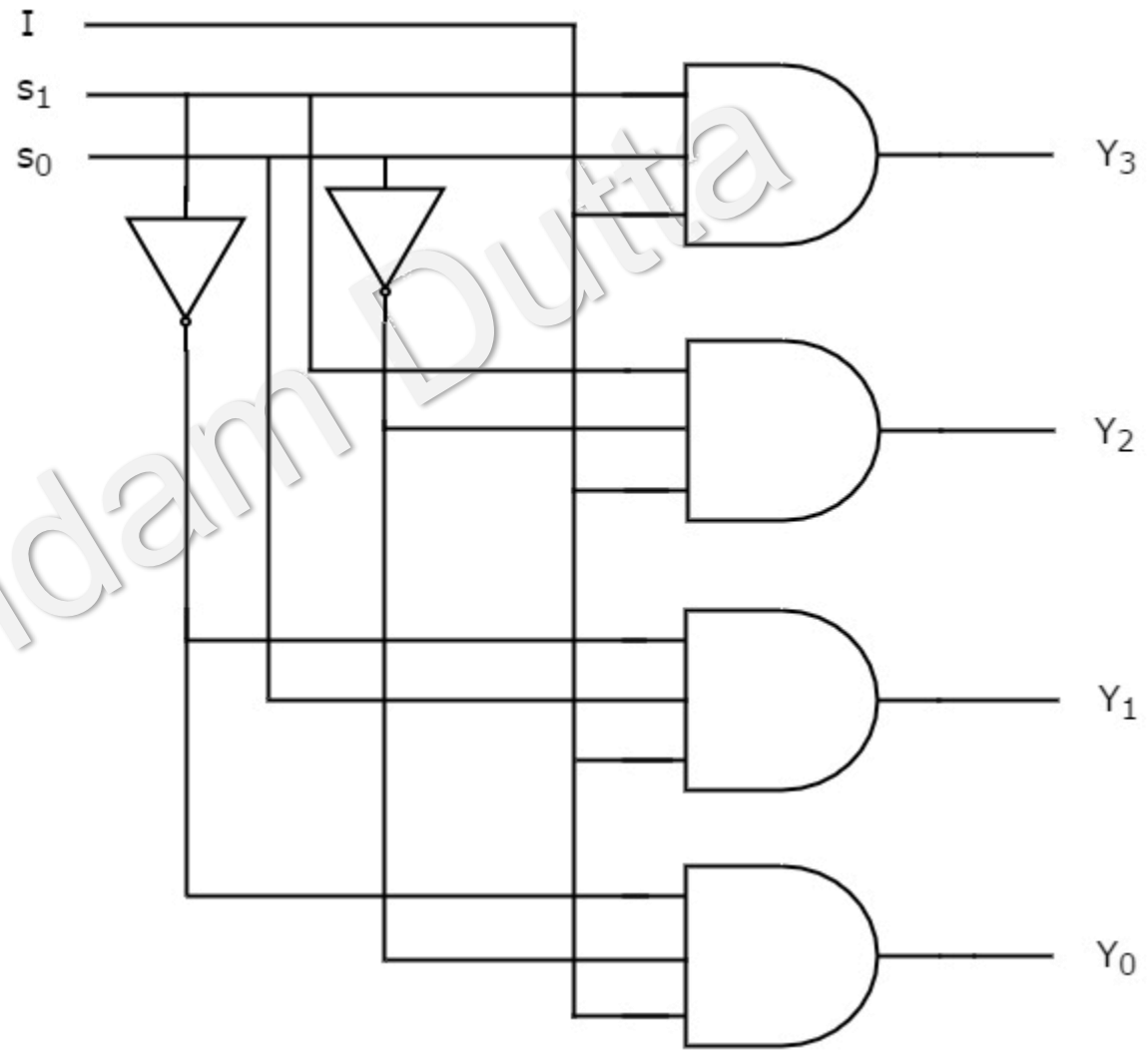
$$Y_0 = \overline{S_1}\,\overline{S_0}I \qquad Y_1 = \overline{S_1}S_0I$$

$$Y_2 = S_1\overline{S_0}I \qquad Y_3 = S_1S_0I$$

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Internal logic circuit of 1:4 DEMUX

S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0



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Design considerations: optimal performance, reliability, cost-effectiveness

Key factors:

1. Number of inputs/outputs
2. Speed/propagation delay
3. Power consumption
4. Noise margin and signal integrity
5. Complexity and cost

Best practices:

1. Systematic approach: carefully evaluate each factor in the context of the specific application to ensure a balanced design

2. Prototype testing: always prototype and test circuits under real-world conditions to validate design decisions

3. Future-proofing: design with potential future needs in mind to avoid costly redesigns

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Differences between MUX and DEMUX

MUX	DEMUX
processes digital information from various sources into a single source	receives digital information from a single source and converts it into several sources
2^n input data lines	single input line
single output data line	2^n output data lines
data selector	data distributor
used at the transmitter end	used at the receiving end

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Encoder

- digital circuit that takes multiple input signals and encodes them into smaller number of output signals
- only one input active at one time
- reduces number of data lines by converting information from one format to another, often by compressing multiple inputs into a binary code

Decoder

- reverse function of encoder: takes a binary input and decodes it into multiple outputs
- interprets the coded information back into a format that can be easily understood or used

Applications: data handling & communication, signal multiplexing/demultiplexing, memory addressing

Encoder: keyboards – encodes a keypress into binary code for the computer to process

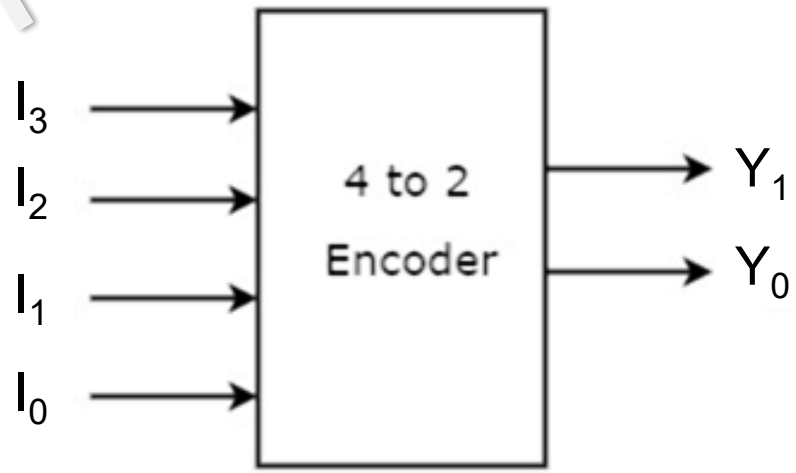
Decoder: digital displays – converts binary coded decimals (BCD) into signals for lighting up appropriate segments

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Encoder:

- typically has 2^n input lines and n output lines
- for e.g., a 4 to 2 encoder has 4 inputs and 2 outputs
- if the inputs are I_0, I_1, I_2, and I_3, then the two outputs will be Y_0 and Y_1

I_3	I_2	I_1	I_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



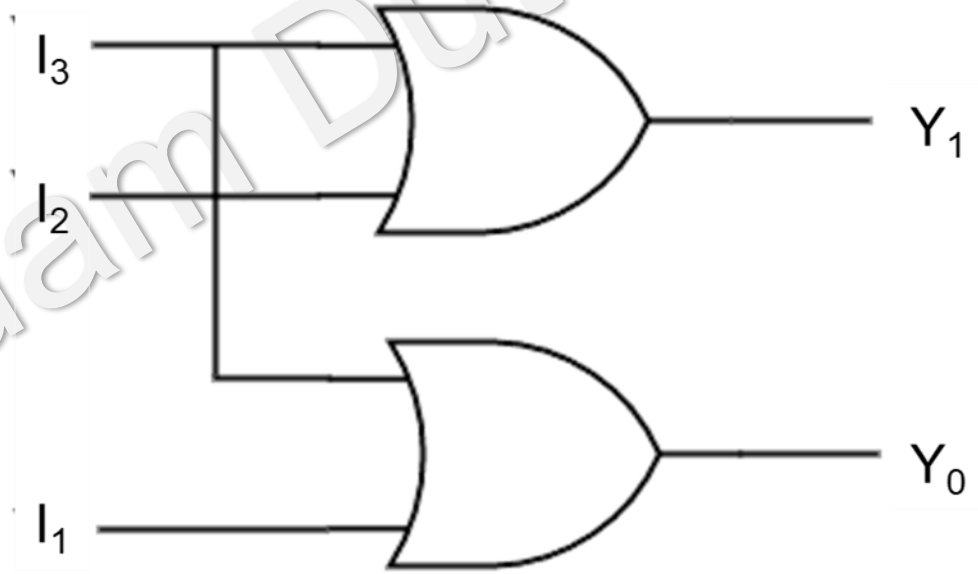
$Y_0 = I_1 + I_3$

$Y_1 = I_2 + I_3$

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Internal logic circuit of 4 to 2 encoder

I_3	I_2	I_1	I_0	Y_1	Y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

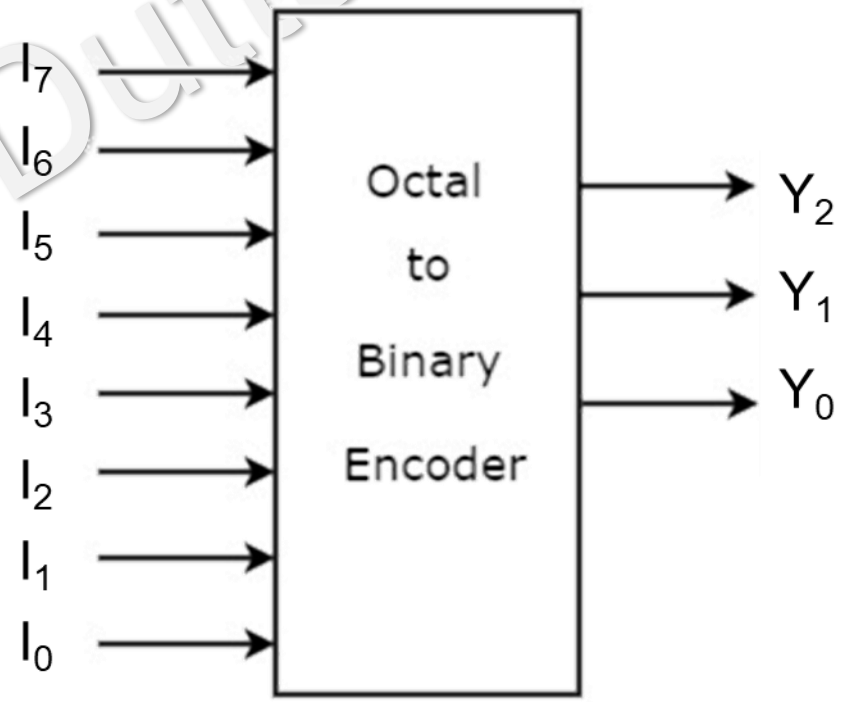


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8 to 3 Encoder (Octal to binary encoder)

- 8 to 3 encoder has 8 inputs and 3 outputs

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1



$Y_0 = I_1 + I_3 + I_5 + I_7$

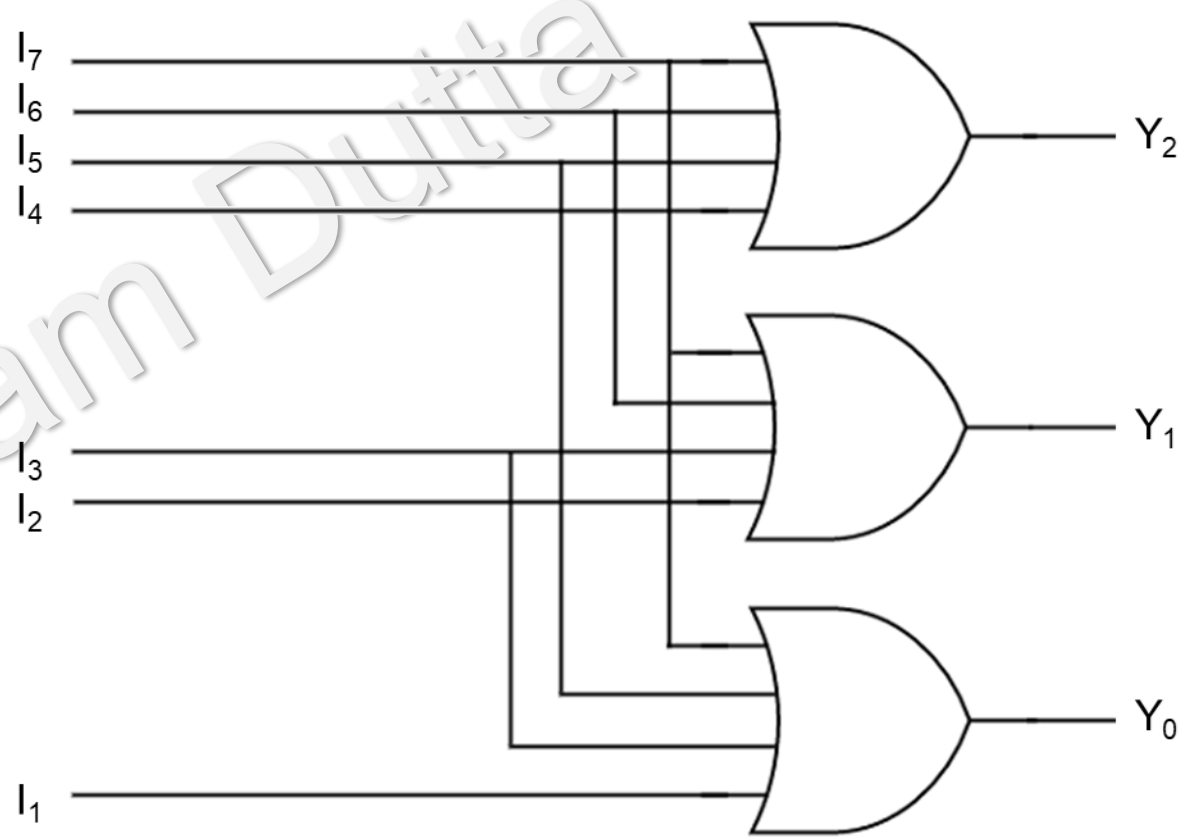
$Y_1 = I_2 + I_3 + I_6 + I_7$

$Y_2 = I_4 + I_5 + I_6 + I_7$

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Internal logic circuit of 8 to 3 encoder

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1



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Decoders:

- typically has 2^n output lines and n input lines
- for e.g., a 2 to 4 decoder has 2 inputs and 4 outputs
- if the inputs are I_0, I_1, then the four outputs will be Y_0, Y_1, Y_2, and Y_3
- one of these outputs will be 1 for each combination of inputs when enable, E is 1

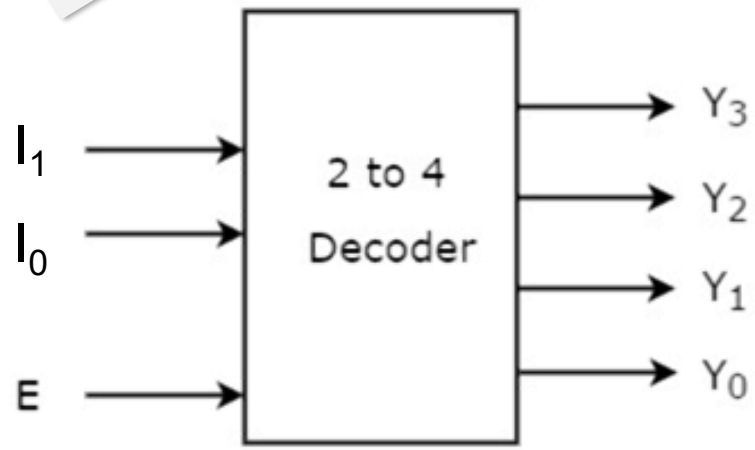
E	I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

$$Y_0 = E\,\overline{I_0}\,\overline{I_1}$$

$$Y_1 = E\,I_0\,\overline{I_1}$$

$$Y_2 = E\,\overline{I_0}\,I_1$$

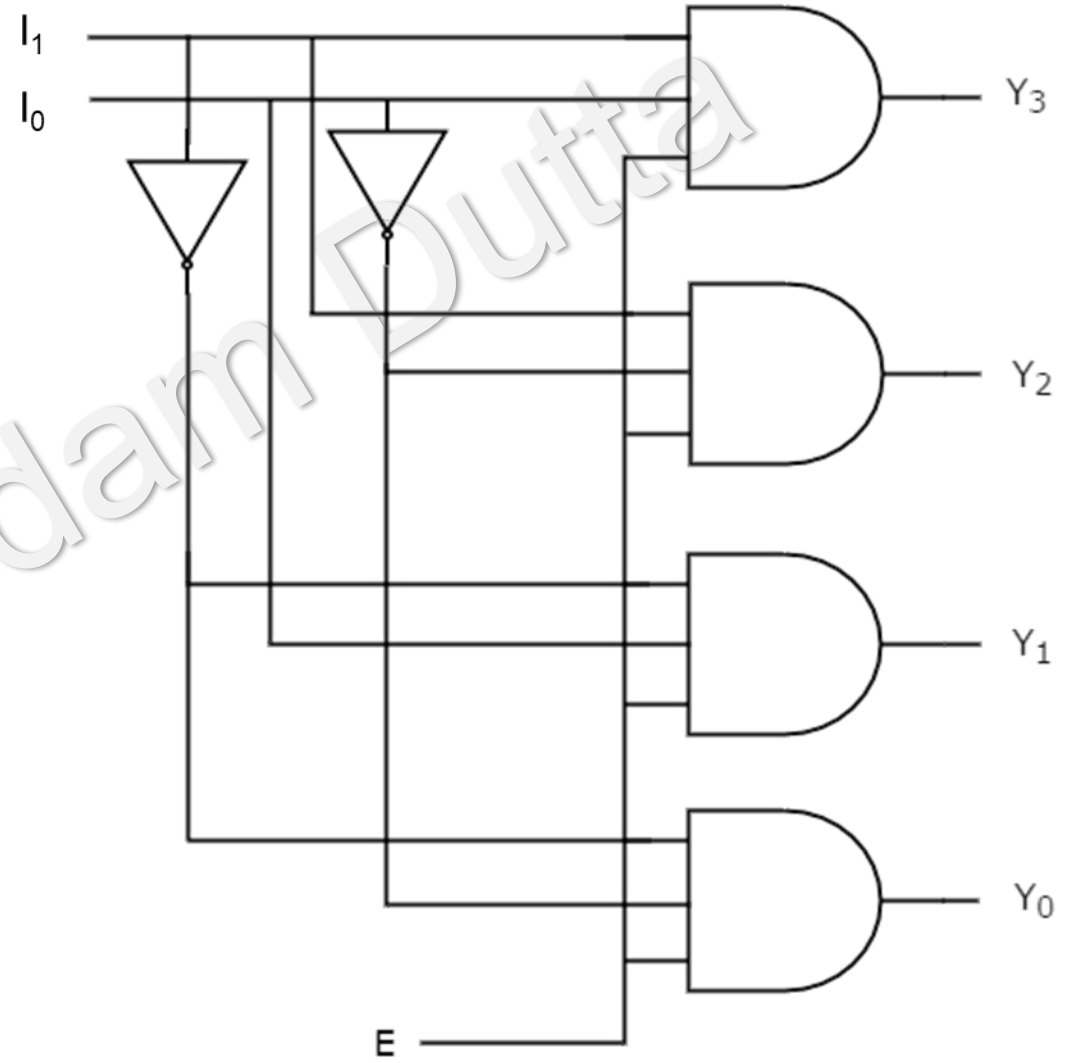
$$Y_3 = E\,I_0\,I_1$$



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Internal logic circuit of 2 to 4 decoder

E	I_1	I_0	Y_3	Y_2	Y_1	Y_0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

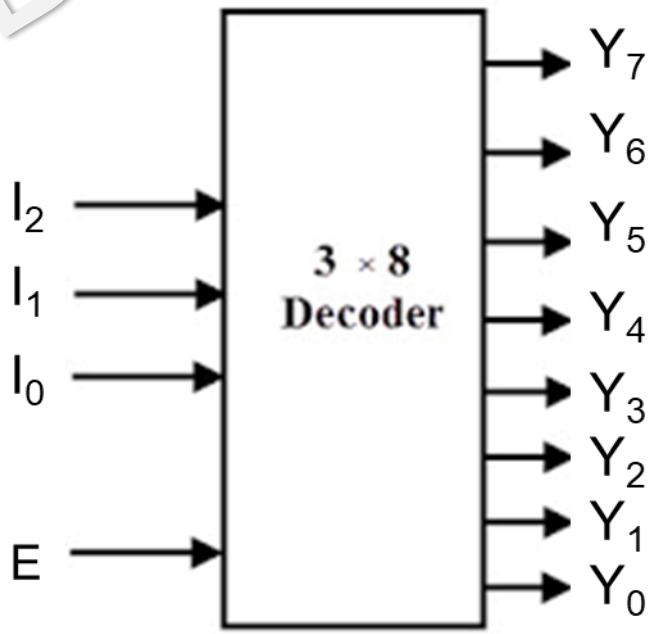


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3 to 8 (Binary to Octal) Decoder:

- if the inputs are I_0, I_1, and I_2 then the eight outputs will be Y_0, Y_1, Y_2, Y_3, Y_4, Y_5, Y_6, and Y_7
- one of these outputs will be 1 for each combination of inputs when enable, E is 1

E	I_2	I_1	I_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0



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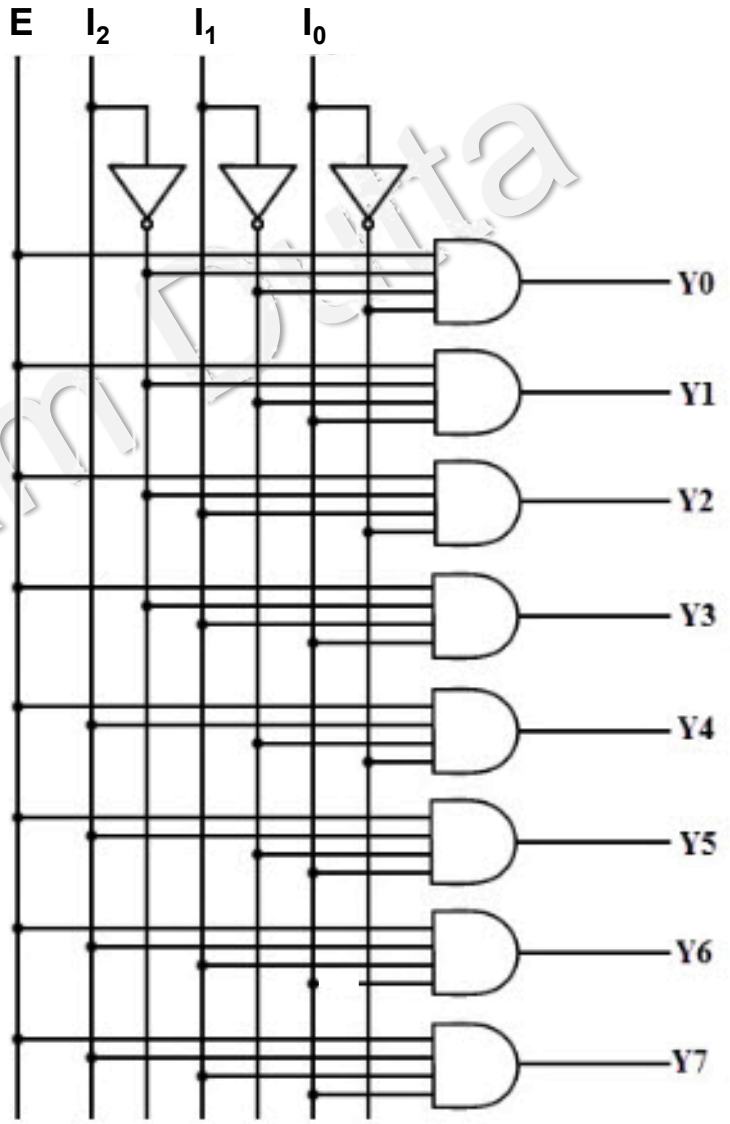
Internal logic circuit of 3 to 8 decoder

$$Y_0 = E\bar{I_2}\bar{I_1}\bar{I_0} \qquad Y_4 = EI_2\bar{I_1}\bar{I_0}$$

$$Y_1 = E\bar{I_2}\bar{I_1}I_0 \qquad Y_5 = EI_2\bar{I_1}I_0$$

$$Y_2 = E\bar{I_2}I_1\bar{I_0} \qquad Y_6 = EI_2I_1\bar{I_0}$$

$$Y_3 = E\bar{I_2}I_1I_0 \qquad Y_7 = EI_2I_1I_0$$



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Design considerations: optimal performance, reliability, cost-effectiveness

Key factors:

1. Number of inputs/outputs
2. Speed/propagation delay
3. Power consumption
4. Noise margin and signal integrity
5. Complexity and cost

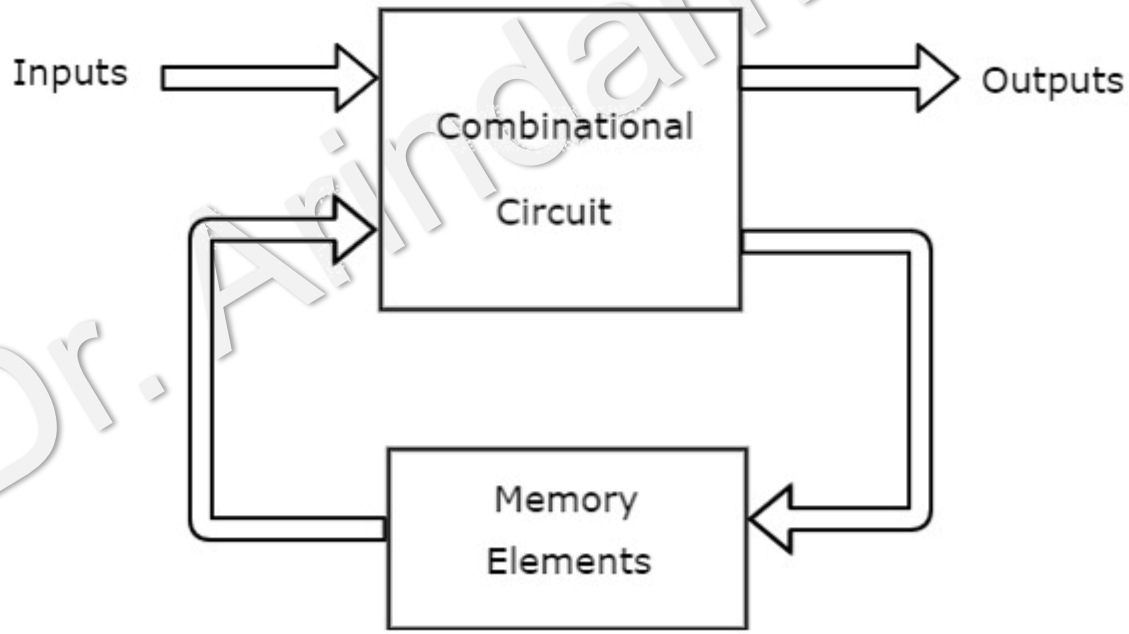
Differences between encoders and decoders

Encoder	Decoder
converts information signal into coded digital bit stream	recovers original information signal from coded bits
number of inputs accepted is 2^n	number of inputs is n
number of outputs is n	number of outputs is 2^n
OR gate is the basic logic element	AND and NOT gates are the basic logic elements
used at the transmitter end	used at the receiving end
applications: e-mail, video encoders, etc.	applications: microprocessors, memory chips, etc.

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Sequential circuit

- consists of a memory element to store history of past operation of the circuit
- output depends on present inputs as well as past outputs of the circuit
- memory element holds the history of circuit operation: latches, flip-flops



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Latch

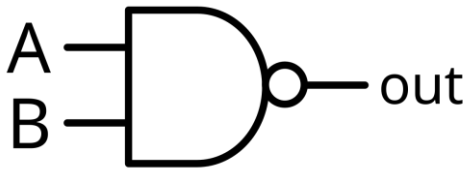
- asynchronous sequential circuit that can store 1-bit information
- two stable states: set and reset
- set state is denoted by 1, reset state is represented by 0
- has feedback mechanism allowing them to maintain the current state until next input is applied

Applications

- used to keep the conditions of the bits to encode binary numbers
- computing as well as data storage
- used in power gating & clock as a storage device

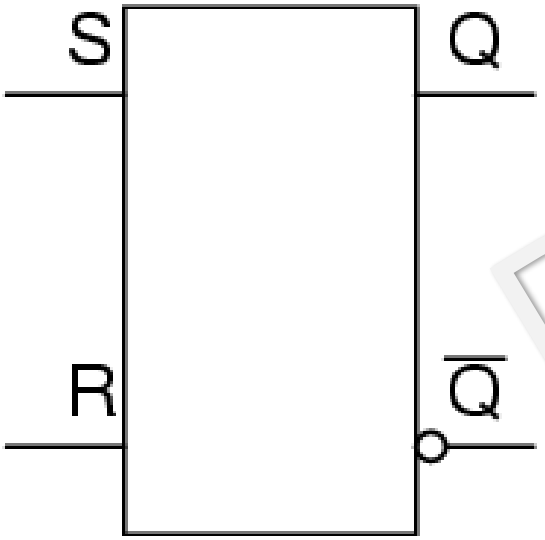
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NAND SR Latch

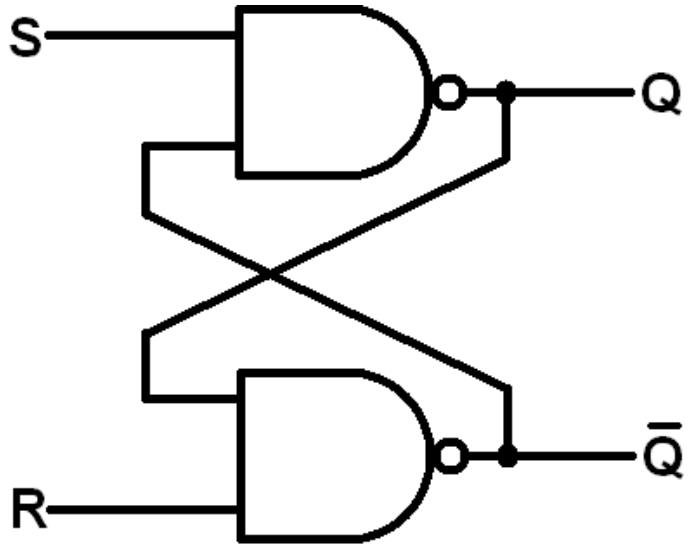


Input	Input	Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

If one input is low (0), output is high (1)



S	R	Q	$\overline{Q}$	State
0	0	x	x	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	Q	$\overline{Q}$	Hold



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Flip-flop

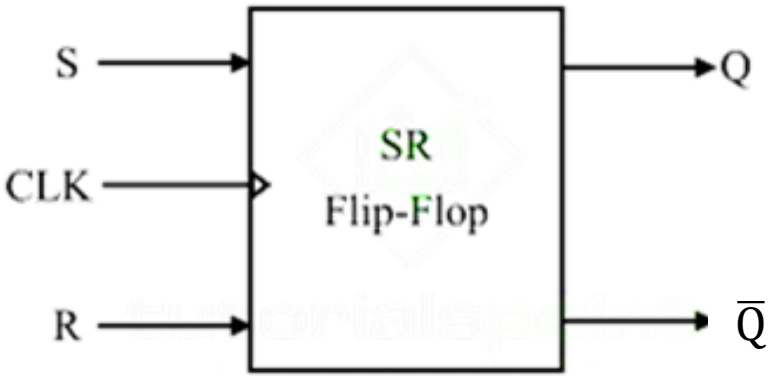
- synchronous or asynchronous sequential digital electronic circuit having two stable states
- used to store one bit of binary data
- fundamental building blocks of all memory devices
- the clock input is used to trigger the flip-flop so that it can change states of its outputs

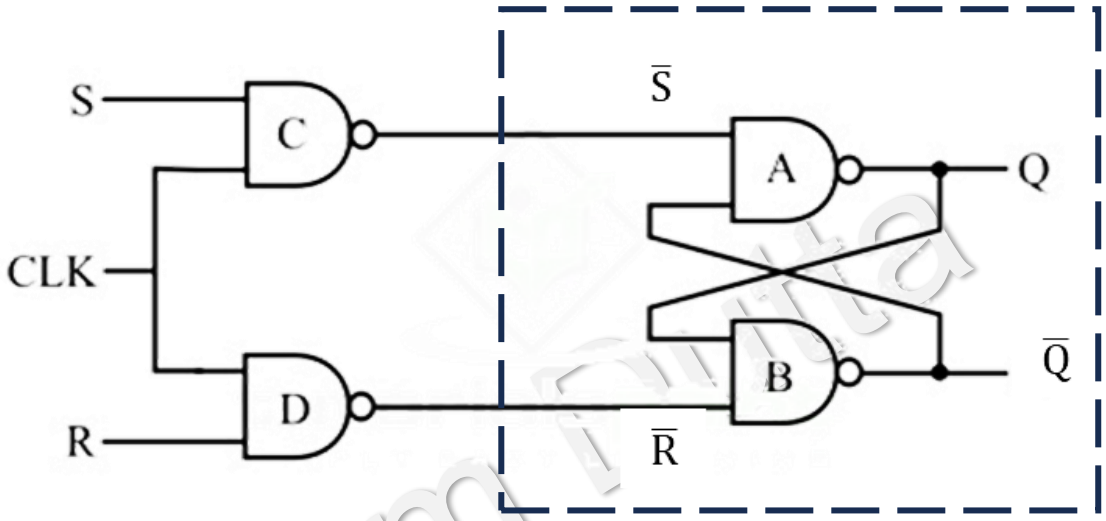
Applications

- frequency dividers
- counters
- storage registers
- shift registers
- data storage
- data transfer

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SR (Set-Reset) flip-flop





NAND SR Latch

- S = 0 and R = 0: output of C and D are S' = 1 and R' = 1. So, outputs of A and B remain unchanged (Hold State/Unchanged)
- S = 0 and R = 1: output of C and D are S' = 1 and R' = 0. So, output of A is 0 and that of B is 1 (Reset State)
- S = 1 and R = 0: output of C and D are S' = 0 and R' = 1. So, output of A is 1 and that of B is 0 (Set State)
- S = 1 and R = 1: output of C and D are S' = 0 and R' = 0. So, outputs of both A and B try to become 1, which is not possible (Invalid State)

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SR flip-flop

Truth Table

Clk	S	R	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	x	Invalid

Characteristic Table

S	R	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	x
1	1	1	x

Excitation Table

Q_n	Q_{n+1}	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0

$$Q_{n+1} = \bar{S}\bar{R}Q_n + S\bar{R}\overline{Q_n} + S\bar{R}Q_n + SR\overline{Q_n} + SRQ_n$$
$$= \bar{S}\bar{R}Q_n + S\bar{R}(\overline{Q_n} + Q_n) + SR(\overline{Q_n} + Q_n)$$
$$= \bar{S}\bar{R}Q_n + S\bar{R} + SR \text{ (Inverse law)}$$
$$= \bar{S}\bar{R}Q_n + S(\bar{R} + R)$$
$$= \bar{S}\bar{R}Q_n + S$$
$$= \bar{S}\bar{R}Q_n + S + \bar{R}Q_n \text{ (Consensus theorem)}$$
$$= S + \bar{R}Q_n(1 + \bar{S})$$
$$= S + \bar{R}Q_n \text{ (Null law)}$$

Dr. Arindam Dutta